

BT401_Genetic Resources & Conservation

Graded Discussion

Board)GDB

Due Date:31 Nov 2024

Fall 2024

Words Limit:Not Given



Genetic Drift

Genetic drift is an evolutionary mechanism that leads to changes in allele frequencies within a population due to random sampling effects. Unlike natural selection, where changes in allele frequencies are driven by an organism's fitness, genetic drift occurs due to chance events that affect which alleles are passed on to the next generation. This random fluctuation in allele frequencies is especially prominent in small populations and can lead to the loss of genetic diversity, the fixation of alleles, or even significant evolutionary changes over time.

In population genetics, genetic drift is a key factor that works alongside mutation, selection, and gene flow. It operates independently of the selective value of alleles and can have profound effects on the genetic structure of a population, particularly in cases where population sizes are small, or "bottlenecks" or "founder effects" occur. Here are the main types of genetic drift:

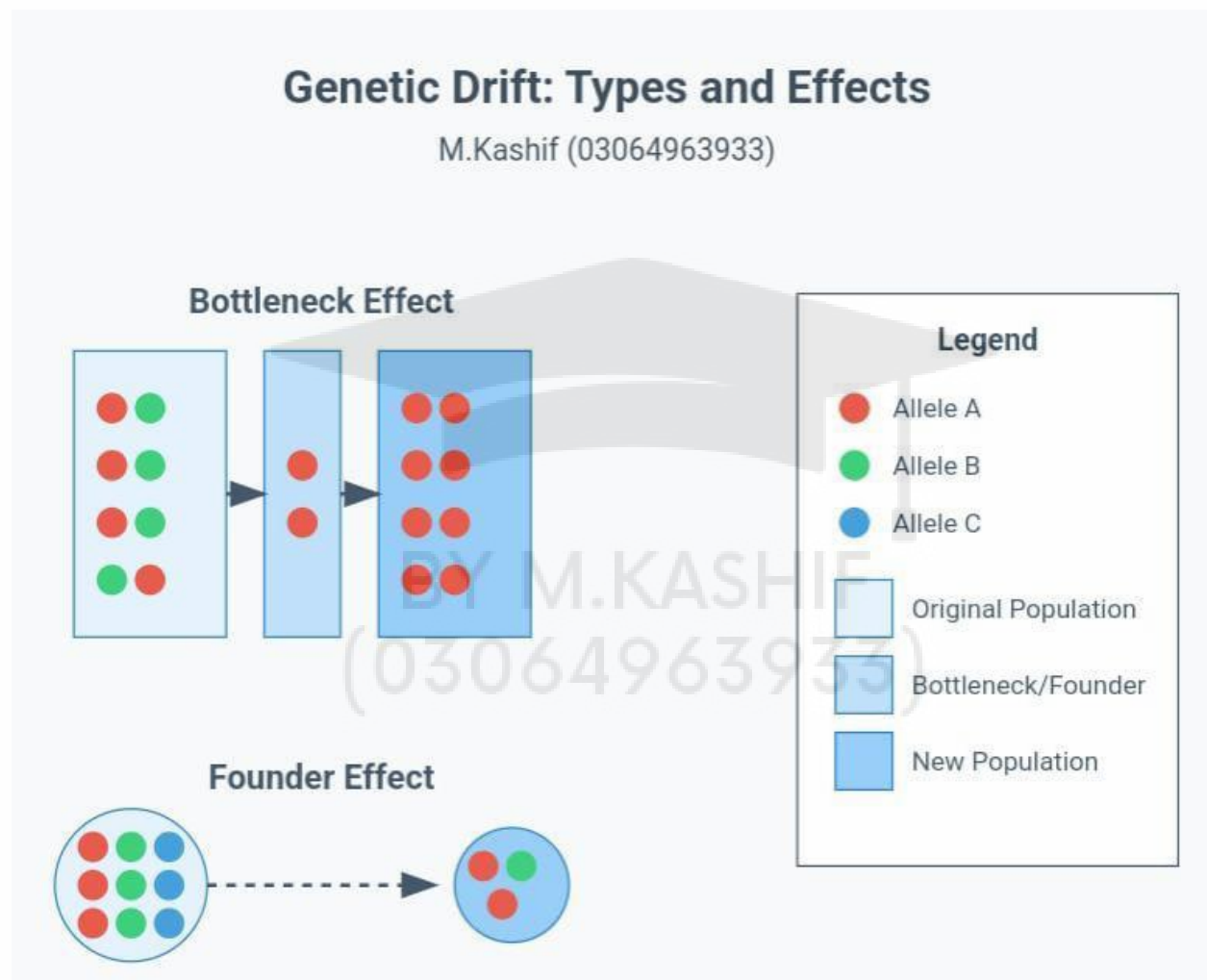
1. Bottleneck Effect

The bottleneck effect occurs when a population undergoes a significant reduction in size due to a random, often catastrophic event such as a natural disaster, disease outbreak, or human intervention (e.g., habitat destruction). This sudden reduction causes a loss of genetic diversity because only a small, random subset of the original population survives. Consequently, the alleles in this subset may not be representative of the original population's genetic diversity. When the

population rebounds, the genetic composition may be significantly altered, often leading to reduced genetic variability and an increased risk of genetic disorders.

Example: The Northern elephant seal population was reduced to just 20 individuals in the late 19th century due to hunting. As the population recovered, it exhibited low genetic variation compared to other seal populations.

2. Founder Effect



The founder effect occurs when a small group from a larger population establishes a new population in a different location. Because the founding group is a small, random sample, it may carry only a subset of the genetic diversity present in the original population. As the new population grows, its genetic composition reflects the alleles of the founding members, often resulting in reduced genetic variation and an increased frequency of certain alleles.

Example: The Amish community in Pennsylvania, originally founded by a small group of European settlers, shows higher frequencies of certain genetic disorders, such as Ellis–van Creveld syndrome, due to the founder effect.

3. Random Fluctuations in Small Populations (Sampling Error)

In small populations, genetic drift is continuously acting due to the randomness associated with allele transmission from one generation to the next. Even in the absence of a bottleneck or founder effect, random sampling in reproduction causes allele frequencies to fluctuate from generation to generation. This type of drift can lead to alleles becoming fixed (frequency reaches 1) or lost (frequency reaches 0) purely by chance. Over time, this can lead to a loss of heterozygosity, where only one allele remains for certain loci within the population.

Example: In a small population of beetles with green and brown coloration alleles, if the green alleles are randomly underrepresented in the offspring generation, the population could drift toward an entirely brown coloration over successive generations.

4. Wright-Fisher Model and the Role of Drift in Evolutionary Dynamics

The Wright-Fisher model provides a framework for understanding genetic drift mathematically. It assumes a finite population size with non-overlapping generations, where each individual's genotype is sampled independently. In this model, genetic drift plays a prominent role in shaping allele frequencies over time, especially in small populations. It suggests that drift can cause allele frequencies to reach fixation or extinction regardless of selective pressure, which can have profound implications on the evolutionary trajectory of small populations.

Significance of Genetic Drift in Evolutionary Theory

Genetic drift can contribute to evolutionary change by altering allele frequencies in populations, especially when acting in concert with other forces like natural selection, mutation, and gene flow. Over long periods, drift can lead to speciation by promoting divergence in isolated populations. For example, in island populations or isolated habitats, drift can produce unique allele compositions that distinguish these populations genetically and phenotypically from their mainland counterparts. This randomness in allele frequency change due to drift contrasts with adaptive evolution and underscores the role of chance in evolution.

Touch at Hearts Emojie to Join our Whatsapp Main Whatsapp Group

